

LAN $\times$ MAN $\leq$ WAN.

LEENA EXERCISE BOOK

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* Answer:-

Q7
→

State Advantages & disadvantages of MAN.

- MAN stands for Metropolitan Area Network

Advantages of MAN:-

i) Less Expensive:

It is less expensive to attach MAN with WAN Network.

ii) Sending local Emails:

You can send local emails fast & free on MAN.

iii) Higher Speed than WAN:

Speed of data can easily reach 1000 Mbps, as MAN uses fiber optics. Files and databases are transferred at fast rates.

iv) Sharing of the Internet:

Users can share their internet connection through installation of MANS. In this way, multiple users can get same high-speed internet.

v) Conversion of LAN to MAN is easy:

MAN is a Combⁿ of 2 or more LAN Networks. So it is faster way to connect 2 LAN networks together.

Disadvantages of MAN:-

i) Difficult to manage:

It is very difficult to manage if the size of network of LANs network increases.

ii) Internet speed difference:

Cannot work on ^{phone} Copper wires. Copper wires affect the speed. So high cost for FO.

iii) Hackers Attack:

There is a high risk of attacking hackers as compared to LAN. So data may be leak.

iv) Technical Staff require to set up:

Highly technical people require to set up MAN.

- (2) **Example :** In the 4B/5B line code 4 data bits are mapped to 5 signal bits to ensure sufficient bit transitions.
- (3) This means that 16 out of the 32 signal possibilities are not used. We can use some reserved signals to indicate the start and end of frames. In effect, we are using "coding violations" to delimit frames.
- (4) In these technique they are reserved signals, it is easy to find the start and end of frames and there is no need to stuff the data. Many data link protocols use a combination of these methods for safety.
- (5) A common pattern used for Ethernet and 802.11 is to have a frame begin with a well-defined pattern called a preamble.
- (6) This pattern might be quite long (72 bits is typical for 802.11) to allow the receiver to prepare for an incoming packet. The preamble is then followed by a length (i.e., count) field in the header that is used to locate the end of the frame.

3.5 ERROR CONTROL

UQ. Explain the purpose of error control from DLL perspective. (MU - Q. 1(c), May 19, 5 Marks)

- (1) Using above technique, the problem of the start and end frame is solved. Another problem is how to make sure that all frames are delivered in proper order to network Layer ?
- (2) In case of unacknowledged connectionless service, the sender just kept outputting frames irrespective of whether they were reached properly.
- (3) But for it is an issue for reliable, connection-oriented service. To ensure the reliable delivery the receiver should send some feedback to sender what exactly happening at the receivers end.
- (4) Typically, in this protocol the receiver has to send back special control frames telling positive or negative acknowledgements about the incoming frames.
- (5) If the sender receives a positive acknowledgement about a frame, it knows the frame has arrived safely.

- (6) If a negative acknowledgement means that something has gone wrong and the frame must be transmitted again.)

What if Frame is Lost Completely or Acknowledge Frame is Lost ?

- **Problem :** If a frame to lost completely (e.g., in a noise burst), then the receiver will not react at all. Also if the acknowledgement frame is lost, the sender will not know how to proceed.
- In above both the cases system will get hanged up if frame is lost with any reason i.e. faulty communication or hardware malfunctioning.
- **Solution :** It is sorted by introducing timers into the data link layer. When the sender transmits a frame, it generally also starts a timer.
- The timer is set to expire after an interval long enough for the frame to reach the destination, be processed there, and have the acknowledgement propagate back to the sender.
- Normally, the frame will be correctly received and the acknowledgement will get back before the timer runs out, in which case the timer will be cancelled.
 - o If either the frame or the acknowledgement is lost, the timer will go off, alerting the sender about a problem.
- Only solution is to transmit the frame again.

Problem : However, when frames may be transmitted multiple times the receiver will accept the same frame two or more times and pass it to the network layer more than once.

- **Solution :** To prevent this from happening, it is generally necessary to assign sequence numbers to outgoing frames, so that the receiver can distinguish retransmissions from originals.

3.6 FLOW CONTROL

UQ. Why is flow control needed ? What are the mechanisms ? (MU - Q. 3(b), Dec. 14, 10 Marks)

UQ. Explain the purpose of flow control from DLL perspective. (MU - Q. 1(c), May 19, 5 Marks)

Explain the purpose of error control from DLL perspective
Ans: From the perspective of data link layer DLL in the OSI model, the purpose of error control is to ensure reliable delivery of data frames.

- (2) Error control in data link layer is a technique used to
- detect errors that may occur during transmission
 - correct errors by automatic retransmission or error correction code

Main purpose of Error Control

- ✓ Error Detection: Enables that the receiver can detect whether the received frame contains any errors.
eg: parity bits, CRC.
- ✓ Error Correction: Enables the receiver to correct errors without needing retransmission using techniques like (hamming code)
- ✓ Reliable Delivery: Ensures that frames are delivered accurately and in sequence. Critical for protocol like TCP/IP which depends on accurate lower layer communication
- ✓ Flow & congestion control: works with flow control to avoid overwhelming the receiver and support efficient work recovery.
- ✓ Frame Acknowledgment: Uses Acknowledgment (ACKs) and negative Acknowledgment (NAKs) to confirm successful delivery or detect lost/corrupted files.

why does the data link protocol always put the CRC in a trailer rather than in header

Ans: In data link layer protocol, the cyclic Redundancy check (CRC) is always placed in the trailer (ie at the end of the frame) rather than in header for following reasons.

(1) CRC Depends on the entire frame content:

- The CRC value is the checksum calculated over the entire content of the frame (typically the header & data fields)
- At that time the header is created, the rest of the frame (especially the data field) might not yet be fully assembled
- Therefore CRC cannot be calculated until entire frame is constructed

(2) Receiver Need to check the whole frame:

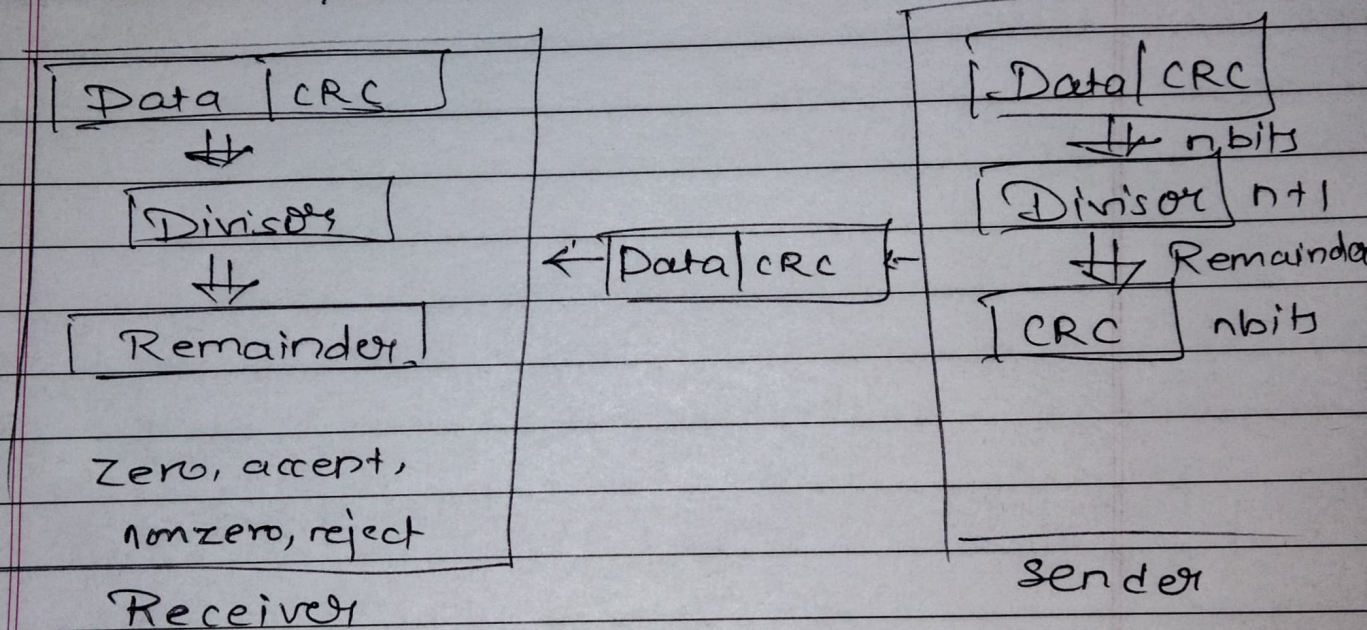
- The receiver reads the entire frame; including data & header & then uses the received CRC to verify integrity.
- Putting CRC at the end ensures that all the bits have been received

(3) Avoids premature validations:

- If the CRC were in the header, the receiver might try to validate the frame before receiving the full content which could result in incorrect validation.

Q. Why does CRC DL protocol always put CRC in trailer (end) rather than in header?
→ frame.

- CRC is a redundancy error technique used to determine error.
- CRC → cyclic Redundancy Check.



UQ. Why does the data link protocol always put the CRC in a trailer rather than in a header ?

(MU – Q. 1(e), May 15, 4 Marks)

- (i) Placing the **CRC** at the end of a frame reduces **packet latency** and reduces hardware buffering requirements.
- (ii) On the transmit side, hardware can read and transmit **bytes of the frame immediately**.
- (iii) The transmitter **calculates the CRC on the fly** as data passes through, then **simply appends the CRC the tail of the frame**.
- (iv) Consider the alternative where the **CRC comes somewhere** in the Ethernet header.
- (v) Hardware must **read and store the entire frame** in order to **calculate the CRC**.
- (vi) This **amounts to a large look-ahead operation** and **adds significantly to transmit latency and hardware cost**.
- (vii) The situation also **becomes more complex** for the receiver as well.



Microwave refers to use of microwave radio waves typically in 1GHz to 300GHz range) to transmit data wirelessly over long distance
eg: Satellite commⁿ

Q.7 State adv & disadv. of microwave.

Adv. of microwave

- 1) They can be used for longer distance
- 2) They can carry 1000's of voice channels at a time.
- 3) High Bandwidth: Microwave freq. can support high data rates
- 4) Quick Deployment: These links are relatively faster & cheaper to deploy compared to Fiber optic cable.
- 5) Require no right of way acquisition betn towers.

Disadv. of microwave

- 1) Reflected from flat surface, like water & metal,
- 2) Weather Sensitivity: Rain, fog, snow can degrade signal quality known as rain fade
- 3) Diffracted around solid objects.
- 4) Limited range without repeaters: Typically 10S microwave links can span around 30-50km; beyond that repeaters or satellite links are needed.

Q.1

Difference betn Subnet, Network & Internetwork.

Parameter	Subnet	Network	Internetwork
Meaning	Consists of Router & communication line.	Consists of Subnet & Host	consists of network of network.
Speed	High	Moderate	Low
msg flow	Short distance	More distance	Longer distance
Area coverage	Smallest part	Limited area	Large geographical area
Example	Part of Network	LAN, MAN, WAN	Internet
Hardware	less than Network	Less H/W	More H/W

Q.2

Explain OSI/ISO model with neat diagram and functions of each layer.

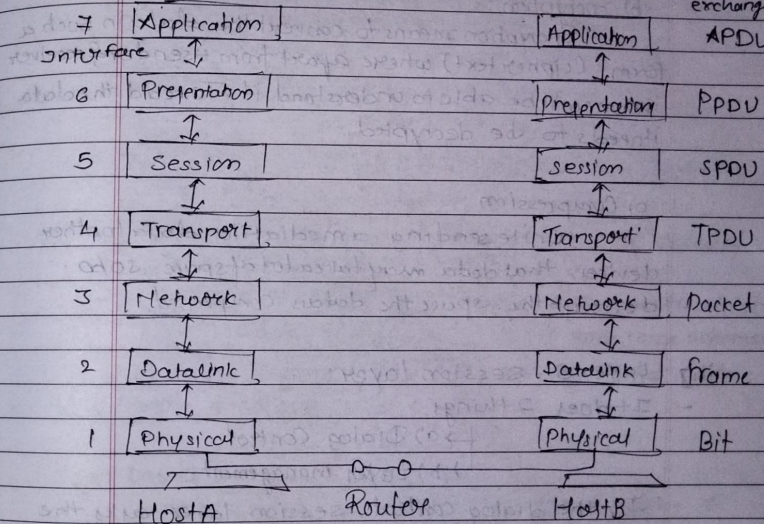
→

- OSI stands for Open System Interconnection is a reference model which was developed by International Organization for Standardization (ISO) in 1984.
- OSI consists of 7 layers & each layer has its own functionality.
- To communicate between 2 devices OSI is used.
- If a sender wants to transfer data to Receiver then it will first go through every layer in its own device

and then it will go through every layer in Receiver device.

- Each layer can interact with its Preceding & succeeding layer.

Name of Unit exchanged.



i) Layer 7 - Application Layer:

- End user layer or Desktop layer a.k.a.
- Acts as an Interface betn system & User.
- Ex. Browser

ii) Layer 6 - Presentation Layer:

- Conversion
 - Encryption
 - Compression
- # C 2

a) Conversion:

converting human understandable language to machine understandable language.

b) Encryption:

Encryption means to convert the data in such a form (ciphertext) where apart from sender & receiver, no one will be able to understand it. To read the data it needs to be decrypted.

c) Compression:

While sending a media file's data to other devices, that data may take a lot of space so to decrease the space the data is compressed.

ii) Layer 5 - Session layer:

- It does 2 things

→ a) Dialog Control

→ b) token management.

In the dialog control, session layer asks the server or if it is free or not to take the data i.e. ask for permission and if it accepts then it is taken to token management.

iii) Transport layer - Layer 4:

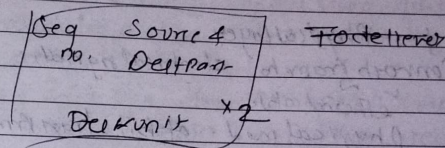
- It does 3 things

a) Segmentation, flow control & error control

↓

Segments.

- The data which is taken from Session layer is divided into multiple units which are also known as Data Units.
- This Data Unit has sequence no. & source & destination port no.



- To get this data delivered in order to receiver, sequence no. of given &.
- These layers can also change the flow & speed of data in server.
- While transmitting data it may happen that data gets corrupted, so at this layer some schemes with the help of which loss or corrupted data gets re-transmitted.

iv) Layer 3 - Network layer

does 2 things, which are logical addressing & path determination.

- This layer assigns an IP address for both sender & receiver and makes a packet.
- It does to make sure the data needed will go to receiver only.
- finds best path to transmit packet. In the no. of routes available.

vi) Layer 2 - Data Link Layer (DLL):

Physical Addressing

error free transfer of data frame.

vii) Layer 1 - Physical Layer

Converts frame to electrical signal.

ethernet cable

physical med. of bit transmission.

Q. Explain any 4 funcⁿ of Data Link Layer with eg.
→ The Data Link layer is 2nd layer in OSI (Open System Interconnection) Model. It's primary role is to ensure reliable data transfer across physical link.

i) Framing:

Provides a way for sender to transmit a set of bits received by receiver, and can be attached special bits pattern to beginning & end of frame.

eg., when computer wants to send a file over a network, the data is too large to be sent in one go. The data link layer breaks the data into smaller frames.

ii) Error Control:

Error Control it detects errors & retransmits damaged or lost frames.

eg., using bits the receiver checks if frame was received correctly. if error is detected it may request

a retransmitting.

iii) Flow Control:

Flow control ensures that fast sender does not overwhelm a slow receiver by sending data too quickly.

eg., If a file transfer betw 2 computers, if sender sends frames faster than receiver can process, receiver may use ACK/NACK (Acknowledgment).

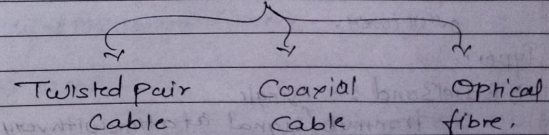
iv) Access Control:

When a single commⁿ channel is shared by multiple devices, a MAC sub-layer of Data Link layer helps to determine which device has right to use medium.

eg., CSMA/CD in Ethernet.

Q. Describe abt the different guided Transmission Media.

→ Guided Transmission Media.



- Guided Transmission media are physical pathways that guide the transmissions of data signals from one device to another, i.e. provide path from 1 device to another.

- Bounded by wire media which include Twisted pair, Coaxial Cable, fiber optic cable.

1) Twisted pair:

- Description: Consists of 2 insulated copper wires twisted together.
- Twisting reduces EMI (Electromagnetic Interference) & cross talk from adj. pairs.

• Types:

i) Unshielded Twisted Pair (UTP)

No extra shielding, cheaper & widely used in Ethernet.

ii) Shielded Twisted Pair (STP)

metallic shielding, to reduce interference better performance in noisy environments.

2) Coaxial Cable:

- Description: Contains 2 conductors parallel to each other. metallic shield (inner/outer) and outer cover.

• Types:-

i) Base Band :- single

transmits signal at a time with very high speed.

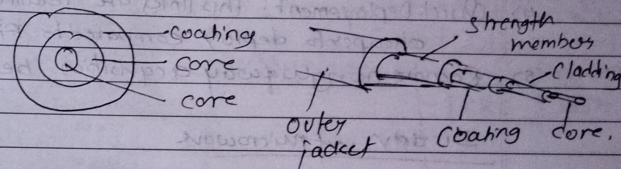
ii) Broad Band :-

transmits simultaneously several signals using different frequencies.

3) Optical fibre cable:

Description:-

- Cable that uses electrical light signals for communication.
- Cable that has 'OF' coated in plastic.
- Immune to electromagnetic interference.
- plastic coating protects from heat, cold.
- 'FO' provides faster data transmission than copper wire.



Current Technology supports 2 modes.

Single mode
Fiber
(SMF)

Multimode
fiber
(MMF)

a) Single MF:-

Very thin core, carries light directly used for longer distance.

b) MMF:-

Thicker core, light reflects multiple times used for shorter distance.